VANISHING OF THE CANONICAL EXT-TWO CLASS CHARACTERIZES LIFTING A BOUNDED-ABOVE COMPLEX ACROSS A SQUARE-ZERO THICKENING

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Editorial note. The task retains the original resolution-based tensor conventions from the canonical-class construction. It does not silently assumeK_0 is a single flat module or perfect.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Sites and sheaves of rings, bounded-above derived categories, flat resolutions, localization of the homotopy category, mapping cones and the four lemma. The canonical class and induced-map support are retained rather than assumed to give the lifting conclusion.

Source reference. Original lemma 0DIT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16953–16953. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05TC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7476–7476. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05QA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7369–7369. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. For every object K_0 in $D^-(\mathcal{O}_0)$ the following are equivalent*

(1) *the class $\omega(K_0) \in \mathrm{Ext}^2_{\mathcal{O}_0}(K_0, K_0 \otimes_{\mathcal{O}_0} \mathcal{I})$ constructed in Lemma 0.2 is zero,*
(2) *there exists $K \in D^-(\mathcal{O})$ with $K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0 = K_0$ in $D(\mathcal{O}_0)$.*

Lemma 0.2. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. For every object K_0 in $D^-(\mathcal{O}_0)$ there is a canonical map*

$$\omega(K_0) : K_0 \longrightarrow K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}[2]$$

Date: stacks-project:defos.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

in $D(\mathcal{O}_0)$ such that for any map $K_0 \to L_0$ in $D^-(\mathcal{O}_0)$ the diagram

$$\begin{array}{ccc} K_0 & \xrightarrow[\omega(K_0)]{} & (K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})[2] \\ \downarrow & & \downarrow \\ L_0 & \xrightarrow{\omega(L_0)} & (L_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})[2] \end{array}$$

commutes.

Proof. Represent K_0 by any complex $\mathcal{K}_0^\bullet$ of $\mathcal{O}_0$-modules. Apply Lemma 0DIT with $\mathcal{G}^n = 0$ for all n. Denote $d : \mathcal{F}^n \to \mathcal{F}^{n+1}$ the maps produced by the lemma. Then we see that $d \circ d : \mathcal{F}^n \to \mathcal{F}^{n+2}$ is zero modulo $\mathcal{I}$. Since $\mathcal{F}^n$ is flat, we see that $\mathcal{I}\mathcal{F}^n = \mathcal{F}^n \otimes_{\mathcal{O}} \mathcal{I} = \mathcal{F}_0^n \otimes_{\mathcal{O}_0} \mathcal{I}$. Hence we obtain a canonical map of complexes

$$d \circ d : \mathcal{F}_0^\bullet \longrightarrow (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2]$$

Since $\mathcal{F}_0^\bullet$ is a bounded above complex of flat $\mathcal{O}_0$-modules, it is K-flat and may be used to compute derived tensor product. Moreover, the map of complexes $\mathcal{F}_0^\bullet \to \mathcal{K}_0^\bullet$ is a quasi-isomorphism by construction. Therefore the source and target of the map just constructed represent K_0 and $K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}[2]$ and we obtain our map $\omega(K_0)$.

Let us show that this procedure is compatible with maps of complexes. Namely, let $\mathcal{L}_0^\bullet$ represent another object of $D^-(\mathcal{O}_0)$ and suppose that

$$\mathcal{K}_0^\bullet \longrightarrow \mathcal{L}_0^\bullet$$

is a map of complexes. Apply Lemma 0DIT for the complex $\mathcal{L}_0^\bullet$, the flat modules $\mathcal{F}^n$, the maps $\mathcal{F}^n \to \mathcal{F}^{n+1}$, and the compositions $\mathcal{F}^n \to \mathcal{K}_0^n \to \mathcal{L}_0^n$ (we apologize for the reversal of letters used). We obtain flat modules $\mathcal{G}^n$, maps $\mathcal{F}^n \to \mathcal{G}^n$, maps $\mathcal{G}^n \to \mathcal{G}^{n+1}$, and maps $\mathcal{G}^n \to \mathcal{L}_0^n$ with all properties as in the lemma. Then it is clear that

$$\begin{array}{ccc} \mathcal{F}_0^\bullet & \longrightarrow & (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \\ \downarrow & & \downarrow \\ \mathcal{G}_0^\bullet & \longrightarrow & (\mathcal{G}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \end{array}$$

is a commutative diagram of complexes.

To see that $\omega(K_0)$ is well defined, suppose that we have two complexes $\mathcal{K}_0^\bullet$ and $(\mathcal{K}_0')^\bullet$ of $\mathcal{O}_0$-modules representing K_0 and two systems $(\mathcal{F}^n, d : \mathcal{F}^n \to \mathcal{F}^{n+1}, \mathcal{F}^n \to \mathcal{K}_0^n)$ and $((\mathcal{F}')^n, d : (\mathcal{F}')^n \to (\mathcal{F}')^{n+1}, (\mathcal{F}')^n \to \mathcal{K}_0^n)$ as above. Then we can choose a complex $(\mathcal{K}_0'')^\bullet$ and quasi-isomorphisms $\mathcal{K}_0^\bullet \to (\mathcal{K}_0'')^\bullet$ and $(\mathcal{K}_0')^\bullet \to (\mathcal{K}_0'')^\bullet$ realizing the fact that both complexes represent K_0 in the derived category. Next, we apply the result of the previous paragraph to

$$(\mathcal{K}_0)^\bullet \oplus (\mathcal{K}_0')^\bullet \longrightarrow (\mathcal{K}_0'')^\bullet$$

This produces a commutative diagram

$$\begin{array}{ccc} \mathcal{F}_0^\bullet \oplus (\mathcal{F}_0')^\bullet & \longrightarrow & (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \oplus ((\mathcal{F}_0')^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \\ \downarrow & & \downarrow \\ \mathcal{G}_0^\bullet & \longrightarrow & (\mathcal{G}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2] \end{array}$$

Since the vertical arrows give quasi-isomorphisms on the summands we conclude the desired commutativity in $D(\mathcal{O}_0)$.

Having established well-definedness, the statement on compatibility with maps is a consequence of the result in the second paragraph. □

Lemma 0.3. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\alpha : K \to L$ be a map of $D^-(\mathcal{O})$. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$-modules. Let $n \in \mathbf{Z}$.*

(1) *If $H^i(\alpha)$ is an isomorphism for $i \geq n$, then $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id_{\mathcal{F}})$ is an isomorphism for $i \geq n$.*
(2) *If $H^i(\alpha)$ is an isomorphism for $i > n$ and surjective for $i = n$, then $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} id_{\mathcal{F}})$ is an isomorphism for $i > n$ and surjective for $i = n$.*

Proof. Choose a distinguished triangle

$$K \to L \to C \to K[1]$$

In case (2) we see that $H^i(C) = 0$ for $i \geq n$. Hence $H^i(C \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}) = 0$ for $i \geq n$ by (the dual of) Derived Categories, Lemma 05TC. This in turn shows that $H^i(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} \mathrm{id}_{\mathcal{F}})$ is an isomorphism for $i > n$ and surjective for $i = n$. In case (1) we moreover see that $H^{n-1}(L) \to H^{n-1}(C)$ is surjective. Considering the diagram

$$\begin{array}{ccc} H^{n-1}(L) \otimes_{\mathcal{O}} \mathcal{F} & \longrightarrow & H^{n-1}(C) \otimes_{\mathcal{O}} \mathcal{F} \\ \downarrow & & \| \\ H^{n-1}(L \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}) & \longrightarrow & H^{n-1}(C \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}) \end{array}$$

we conclude the lower horizontal arrow is surjective. Combined with what was said before this implies that $H^n(\alpha \otimes^{\mathbf{L}}_{\mathcal{O}} \mathrm{id}_{\mathcal{F}})$ is an isomorphism. □

Proof. Let K be as in (2). Then we can represent K by a bounded above complex $\mathcal{F}^\bullet$ of flat $\mathcal{O}$-modules. Then $\mathcal{F}_0^\bullet = \mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{O}_0$ represents K_0 in $D(\mathcal{O}_0)$. Since $d_{\mathcal{F}^\bullet} \circ d_{\mathcal{F}^\bullet} = 0$ as $\mathcal{F}^\bullet$ is a complex, we see from the very construction of $\omega(K_0)$ that it is zero.

Assume (1). Let $\mathcal{F}^n$, $d : \mathcal{F}^n \to \mathcal{F}^{n+1}$ be as in the construction of $\omega(K_0)$. The nullity of $\omega(K_0)$ implies that the map

$$\omega = d \circ d : \mathcal{F}_0^\bullet \longrightarrow (\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I})[2]$$

is zero in $D(\mathcal{O}_0)$. By definition of the derived category as the localization of the homotopy category of complexes of $\mathcal{O}_0$-modules, there exists a quasi-isomorphism $\alpha : \mathcal{G}_0^\bullet \to \mathcal{F}_0^\bullet$ such that there exist $\mathcal{O}_0$-modules maps $h^n : \mathcal{G}_0^n \to \mathcal{F}_0^{n+1} \otimes_{\mathcal{O}} \mathcal{I}$ with

$$\omega \circ \alpha = d_{\mathcal{F}_0^\bullet \otimes \mathcal{I}} \circ h + h \circ d_{\mathcal{G}_0^\bullet}$$

We set

$$\mathcal{H}^n = \mathcal{F}^n \times_{\mathcal{F}_0^n} \mathcal{G}_0^n$$

and we define

$$d' : \mathcal{H}^n \longrightarrow \mathcal{H}^{n+1}, \quad (f^n, g_0^n) \longmapsto (d(f^n) - h^n(g_0^n), d(g_0^n))$$

with obvious notation using that $\mathcal{F}_0^{n+1} \otimes_{\mathcal{O}_0} \mathcal{I} = \mathcal{F}^{n+1} \otimes_{\mathcal{O}} \mathcal{I} = \mathcal{I}\mathcal{F}^{n+1} \subset \mathcal{F}^{n+1}$. Then one checks $d' \circ d' = 0$ by our choice of h^n and definition of ω. Hence $\mathcal{H}^\bullet$ defines an object in $D(\mathcal{O})$. On the other hand, there is a short exact sequence of complexes of $\mathcal{O}$-modules

$$0 \to \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} \to \mathcal{H}^\bullet \to \mathcal{G}_0^\bullet \to 0$$

We still have to show that $\mathcal{H}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$ is isomorphic to K_0. Choose a quasi-isomorphism $\mathcal{E}^\bullet \to \mathcal{H}^\bullet$ where $\mathcal{E}^\bullet$ is a bounded above complex of flat $\mathcal{O}$-modules. We obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I} & \longrightarrow & \mathcal{E}^\bullet & \longrightarrow & \mathcal{E}_0^\bullet & \longrightarrow & 0 \\ & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \\ 0 & \longrightarrow & \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} & \longrightarrow & \mathcal{H}^\bullet & \longrightarrow & \mathcal{G}_0^\bullet & \longrightarrow & 0 \end{array}$$

We claim that δ is a quasi-isomorphism. Since $H^i(\delta)$ is an isomorphism for $i \gg 0$, we can use descending induction on n such that $H^i(\delta)$ is an isomorphism for $i \geq n$. Observe that $\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}$ represents $\mathcal{E}_0^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}$, that $\mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I}$ represents $\mathcal{G}_0^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I}$, and that $\beta = \delta \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathrm{id}_{\mathcal{I}}$ as maps in $D(\mathcal{O}_0)$. This is true because $\beta = (\alpha \otimes \mathrm{id}_{\mathcal{I}}) \circ (\delta \otimes \mathrm{id}_{\mathcal{I}})$. Suppose that $H^i(\delta)$ is an isomorphism in degrees $\geq n$. Then the same is true for β by what we just said and Lemma 0.3. Then we can look at the diagram

$$\begin{array}{ccccccccc}
H^{n-1}(\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^{n-1}(\mathcal{E}^\bullet) & \longrightarrow & H^{n-1}(\mathcal{E}_0^\bullet) & \longrightarrow & H^n(\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^n(\mathcal{E}^\bullet) \\
\downarrow{\scriptstyle H^{n-1}(\beta)} & & \downarrow & & \downarrow{\scriptstyle H^{n-1}(\delta)} & & \downarrow{\scriptstyle H^n(\beta)} & & \downarrow \\
H^{n-1}(\mathcal{F}_0^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^{n-1}(\mathcal{H}^\bullet) & \longrightarrow & H^{n-1}(\mathcal{G}_0^\bullet) & \longrightarrow & H^n(\mathcal{F}_0^\bullet \otimes_{\mathcal{O}} \mathcal{I}) & \longrightarrow & H^n(\mathcal{H}^\bullet)
\end{array}$$

Using Homology, Lemma 05QA we see that $H^{n-1}(\delta)$ is surjective. This in turn implies that $H^{n-1}(\beta)$ is surjective by Lemma 0.3. Using Homology, Lemma 05QA again we see that $H^{n-1}(\delta)$ is an isomorphism. The claim holds by induction, so δ is a quasi-isomorphism which is what we wanted to show. □

References